

Jason Teixeira

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Professional Summary

Senior Quantitative Engineer with 5+ years building high-performance trading systems, cloud infrastructure, and automated testing frameworks. Expert in algorithmic trading, ML/AI, distributed systems, and DevOps. Proven track record deploying production systems processing millions in trading volume with Kubernetes orchestration and event-driven architectures.

Technical Skills

Languages: Python, JavaScript, TypeScript, Go, Bash, Shell, SQL, HTML/CSS, C++, Java, Rust, R

Web Frameworks: Next.js 15, React 19, React 18, Vue.js, Angular, Node.js, Express, FastAPI, Flask, Django, Svelte, Remix

Python Libraries: pandas, numpy, scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, matplotlib, seaborn, plotly, statsmodels

Cloud Platforms: AWS (EC2, S3, Lambda, RDS, DynamoDB, CloudFormation, ECS, EKS, CloudWatch, IAM, API Gateway, SQS, SNS), Azure, GCP

DevOps & Infrastructure: Kubernetes, Docker, Terraform, Ansible, Helm, ArgoCD, Jenkins, GitLab CI/CD, GitHub Actions, CircleCI, Pulumi

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch, Cassandra, TimescaleDB, InfluxDB, SQLite

Message Queues & Streaming: Kafka, RabbitMQ, Apache Pulsar, AWS SQS/SNS, Redis Streams, NATS

Testing & QA: pytest, unittest, Jest, Mocha, Selenium, Playwright, Cypress, Postman, JMeter, Locust, k6, Robot Framework, TestNG

Monitoring & Observability: Prometheus, Grafana, CloudWatch, Datadog, New Relic, ELK Stack, Jaeger, OpenTelemetry, PagerDuty

Data Engineering: Apache Spark, Apache Airflow, Apache Flink, Dask, Ray, Luigi, Prefect, dbt, Snowflake

Finance Technologies: Algorithmic Trading, HFT Systems, Risk Analytics (VaR, CVaR, Greeks), Market Microstructure, Portfolio Optimization, Options Pricing, Backtesting Engines, Order Management Systems (OMS), FIX Protocol, Bloomberg API, Interactive Brokers API

Machine Learning: Deep Learning, Reinforcement Learning, NLP, Computer Vision, Time Series Forecasting, Feature Engineering, Model Deployment, AutoML, MLOps

Version Control & Collaboration: Git, GitHub, GitLab, Bitbucket, Jira, Confluence, Notion, Linear

API Development: REST, GraphQL, gRPC, WebSocket, OpenAPI/Swagger, OAuth2, JWT Authentication

Security & Compliance: OWASP, SSL/TLS, Encryption, IAM, RBAC, Vulnerability Scanning, Penetration Testing, SOC2, GDPR

Professional Experience

Senior Quantitative Engineer | HighStrike | Remote

Jan 2020 – Present

Architected and deployed production algorithmic trading systems processing \$10M+ daily volume using Python, Kubernetes, and event-driven microservices. Built ML-driven alpha generation platform with real-time signal processing achieving 2.45 Sharpe ratio in backtesting. Designed comprehensive risk analytics platform with VaR, CVaR, Greeks calculation, and Monte Carlo simulations. Implemented HFT market microstructure analytics with order book reconstruction and microsecond-precision tick data processing. Deployed Kubernetes-orchestrated infrastructure with Ray/Dask distributed computing, Kafka event streaming, and auto-scaling based on market conditions. Developed full-stack trading platform using Next.js 15, React 19, TypeScript with 590K+ LOC featuring admin dashboard with 60+ features, WebSocket integration, portfolio tracking, and JWT authentication with RBAC.

Senior Systems Engineer | The Home Depot | Atlanta, GA

Aug 2018 – Dec 2021

Led enterprise cloud migration from on-premise to AWS, architecting scalable infrastructure using Terraform, CloudFormation, and Kubernetes. Reduced infrastructure costs by 35% through optimization of EC2 instances, S3 storage policies, and Lambda functions. Built CI/CD pipelines using Jenkins and GitLab reducing deployment time from 4 hours to 15 minutes. Implemented comprehensive monitoring with CloudWatch, Prometheus, and Grafana for 99.9% uptime SLA. Developed automated testing frameworks using Selenium, pytest, and Postman for regression testing across 50+ microservices.

Systems Specialist | The Home Depot | Atlanta, GA

Jan 2015 – Aug 2018

Automated infrastructure provisioning using Python, Bash, and Ansible reducing setup time by 70%. Implemented monitoring and alerting systems using CloudWatch and PagerDuty. Maintained 99.5% uptime for critical e-commerce systems serving 2M+ daily users. Created technical documentation and runbooks for operations team.

IT Support Specialist | The Home Depot | Atlanta, GA

Jun 2012 – Jan 2015

Provided technical support for 500+ employees across retail and corporate environments. Resolved hardware, software, and network issues maintaining 95% satisfaction rating. Deployed and configured workstations, managed Active Directory, and assisted with system upgrades.

Education

Full Sail University | Bachelor of Science, Computer Science | 2008 – 2012

Kean University | Bachelor of Science, Finance | 2008 – 2012

Certifications & Awards

AWS Certified Solutions Architect – Associate | Amazon Web Services

AWS Certified Developer – Associate | Amazon Web Services

Performance Recognition: Reduced infrastructure costs by 35% at Home Depot, optimized trading systems achieving 2.45 Sharpe ratio

Selected Projects

QuantumTrader – Elite quantitative finance portfolio with production-grade algorithmic trading system achieving 2.45 Sharpe ratio and 42.7% annual return. Comprehensive risk analytics platform (RiskLab) with VaR, CVaR, Greeks calculation, Monte Carlo simulations, and portfolio optimization using modern portfolio theory and Black-Scholes models. ML/AI trading platform (AlphaForge) with deep learning models (LSTM, Transformers), reinforcement learning agents (DQN, PPO), and 100+ feature engineering indicators including technical analysis, sentiment analysis, and alternative data sources. HFT market microstructure analytics with order book reconstruction, liquidity analysis, VWAP/TWAP calculations, and tick data processing at microsecond precision. Kubernetes-orchestrated cloud infrastructure with distributed computing (Ray, Dask), event-driven architecture (Kafka), Redis caching, TimescaleDB for time-series data, and auto-scaling based on market volatility. Stack: Python, TensorFlow, PyTorch, pandas, numpy, Kafka, Kubernetes, PostgreSQL, Redis, FastAPI.

NexQuantSite – Production-grade Next.js 15 trading platform with 590K+ LOC featuring comprehensive admin panel with 60+ features including real-time analytics dashboard, user management system with RBAC, bot performance monitoring with P&L tracking, risk management console, financial operations center, trade execution monitoring, and system health metrics. Real-time trading interface with WebSocket integration for live price feeds, portfolio tracking with real-time P&L calculations, automated trading bots with strategy configuration, signal generation with ML predictions, order management system, and multi-account support. Authentication system with JWT + refresh tokens, RBAC permissions engine, session management using Zustand, OAuth2 integration, 2FA support, and audit logging. Performance optimization achieved through code splitting, dynamic imports, lazy loading, React Server Components, edge caching, and bundle size reduction from 2MB to <500KB. Stack: Next.js 15, React 19, TypeScript, Tailwind CSS, Zustand, tRPC, Prisma, PostgreSQL, WebSocket, Redis.

AlphaStream – ML-driven trading signals platform with predictive alpha generation using ensemble methods (Random Forest, XGBoost, LightGBM), feature engineering pipeline processing 500+ technical indicators, backtesting engine with walk-forward optimization, and out-of-sample validation. Real-time market data processing with sub-second latency, signal distribution via WebSocket and REST APIs, and performance analytics dashboard tracking Sharpe ratio, drawdown, and win rate. Integrated with multiple broker APIs (Interactive Brokers, Alpaca, TD Ameritrade) for automated execution with risk controls and position sizing algorithms. Model retraining pipeline with MLflow tracking and A/B testing framework. Stack: Python, scikit-learn, XGBoost, FastAPI, Redis, Celery, PostgreSQL, Docker, Kubernetes.

E-Commerce Test Suite – Comprehensive automated testing framework for e-commerce platforms using Selenium WebDriver, pytest framework, and Page Object Model design pattern. Covers end-to-end user flows including product search and filtering, cart management, multi-step checkout process, payment gateway integration (Stripe, PayPal), inventory management validation, order processing workflows, and admin panel operations. Integrated with Jenkins CI/CD pipeline for automated regression testing on every deployment achieving 95% code coverage. Features include parallel test execution across multiple browsers (Chrome, Firefox, Safari), screenshot capture on failures, detailed HTML reports with test metrics, and Slack notifications. Data-driven testing with parameterization for testing multiple user scenarios and edge cases. Stack: Python, Selenium, pytest, Jenkins, Docker, Allure Reports.

Web Automation Test Framework – Enterprise-grade Selenium-based test automation framework with parallel execution using pytest-xdist, cross-browser testing supporting Chrome, Firefox, Edge, and Safari, cloud testing integration with BrowserStack and Sauce Labs, and detailed reporting with screenshots, videos, and execution metrics. Reduced manual testing time by 80% through automated regression suites covering critical business workflows including user authentication, data validation, form submissions, and complex multi-step processes. Features include dynamic waits, custom exceptions, retry mechanisms, test data management, configuration-driven test execution, and integration with Jira for bug tracking. Stack: Python, Selenium, pytest, BrowserStack, Jenkins, Allure, Docker.

API Test Automation Wireframe – REST API testing framework using pytest, requests library, and JSON schema validation with comprehensive test coverage for 50+ API endpoints. Supports multiple authentication methods (API keys, OAuth2, JWT), data-driven testing with CSV/JSON test data, request/response validation, performance testing with response time assertions, and contract testing with OpenAPI/Swagger specifications. Integrated with GitLab CI/CD pipelines for continuous testing on every code commit. Features include environment-specific configurations, test data factories, custom assertions, detailed logging, and HTML reports with request/response details. Handles complex scenarios including file uploads, pagination, rate limiting, and error handling. Stack: Python, pytest, requests, JSON Schema, GitLab CI, Docker, Newman.

Additional Projects & Contributions

Real-Time Risk Management Dashboard – Built enterprise risk monitoring system with real-time portfolio analytics, VaR calculations updated every 60 seconds, position limit monitoring with automated alerts, margin requirement tracking, and exposure analysis across asset classes. Implemented WebSocket-based live data feeds from multiple exchanges, Redis caching for sub-millisecond query performance, and PostgreSQL for historical data storage. Dashboard features include customizable risk metrics, P&L attribution analysis, scenario modeling, and stress testing capabilities. Stack: Python, FastAPI, React, WebSocket, Redis, PostgreSQL, Grafana.

Distributed Data Pipeline for Market Data – Architected high-throughput data ingestion pipeline processing 500K+ market events per second from multiple exchanges (NYSE, NASDAQ, CME, ICE). Implemented Kafka-based streaming architecture with partitioning strategy for parallel processing, Spark Structured Streaming for real-time aggregations, and TimescaleDB for time-series storage with automatic data retention policies. Built data quality monitoring with automated anomaly detection, data lineage tracking, and comprehensive alerting. Reduced data processing latency from 5 seconds to <100ms. Stack: Python, Apache Kafka, Apache Spark, TimescaleDB, Airflow, Kubernetes.

Machine Learning Model Deployment Platform – Developed MLOps platform for automated model training, evaluation, and deployment with A/B testing framework. Features include model versioning with MLflow, automated retraining pipelines triggered by model drift detection, feature store for consistent feature engineering, model serving with FastAPI and TensorFlow Serving, monitoring dashboards tracking model performance metrics, and rollback capabilities. Deployed 25+ ML models serving 1M+ predictions daily with p99 latency <100ms. Stack: Python, MLflow, TensorFlow, FastAPI, Docker, Kubernetes, Prometheus, Grafana.

Cloud Infrastructure Automation – Built Infrastructure-as-Code platform managing 200+ AWS resources across multiple environments (dev, staging, prod). Implemented Terraform modules for VPC networking, EKS clusters, RDS databases, S3 buckets with versioning and encryption, CloudFront CDN, and Lambda functions. Created GitOps workflow with ArgoCD for continuous deployment, automated terraform plan/apply in CI/CD pipelines, drift detection and remediation, cost monitoring with budget alerts, and comprehensive tagging strategy for resource organization. Stack: Terraform, AWS, Kubernetes, ArgoCD, GitLab CI, Python.

Performance Testing Framework – Developed comprehensive load testing framework using Locust and k6 for stress testing trading systems under peak market conditions. Simulates 10K+ concurrent users with realistic trading patterns, measures response times across percentiles (p50, p95, p99), identifies bottlenecks through distributed tracing, and generates detailed performance reports. Automated performance regression testing in CI/CD pipeline catching performance degradations before production. Helped optimize system to handle 3x traffic during market volatility. Stack: Python, Locust, k6, Grafana, InfluxDB, Docker.

Security Hardening & Compliance Automation – Implemented security scanning pipeline with automated vulnerability detection using Trivy, OWASP ZAP, and Snyk. Built compliance framework for SOC2 requirements including automated security controls, audit logging, access reviews, encryption at rest and in transit, and secrets management with HashiCorp Vault. Integrated security gates in CI/CD preventing deployment of vulnerable code. Achieved 95% reduction in security findings through automated remediation. Stack: Docker, Trivy, OWASP ZAP, Vault, AWS KMS, GitLab CI.

Technical Expertise Highlights

Microservices Architecture: Designed and deployed distributed systems with 50+ microservices using service mesh (Istio), API Gateway patterns, event-driven communication, circuit breakers, and distributed tracing.

Performance Optimization: Reduced system latency from 500ms to <50ms through query optimization, caching strategies (Redis, CDN), database indexing, connection pooling, and async processing.

Scale & Reliability: Built systems handling 10M+ daily transactions with 99.9% uptime, auto-scaling capabilities, load balancing, fault tolerance, and disaster recovery procedures.

Cost Optimization: Reduced cloud infrastructure costs by 35% through reserved instances, spot instances, S3 lifecycle policies, Lambda optimization, and resource rightsizing.

Team Leadership: Mentored junior engineers, conducted code reviews, established engineering best practices, created technical documentation, and led architecture design sessions.

Algorithm Optimization: Optimized trading algorithms reducing execution time from $O(n^2)$ to $O(n \log n)$, implemented efficient data structures (heaps, trees, hash maps), and parallel processing achieving 10x performance improvement.

Database Performance: Optimized database queries reducing response time by 95%, implemented connection pooling, read replicas for scaling reads, partitioning strategies for large tables, and materialized views for complex aggregations.

Real-Time Systems: Built low-latency systems with microsecond precision timing, lock-free data structures, memory-mapped files, zero-copy networking, and CPU pinning for consistent performance.

Data Science: Applied statistical analysis, hypothesis testing, time series forecasting (ARIMA, Prophet), anomaly detection, causal inference, and experimentation frameworks for A/B testing.

DevSecOps: Integrated security throughout SDLC with automated vulnerability scanning, SAST/DAST tools, secrets management, least-privilege IAM policies, and compliance automation.

Open Source & Community

Contributions: Active contributor to open-source projects in Python ecosystem, trading libraries, and DevOps tools. Published technical articles on algorithmic trading, cloud architecture, and testing best practices.

Speaking & Knowledge Sharing: Presented technical talks on microservices architecture, ML model deployment, and performance optimization. Mentored engineers through technical workshops and code review sessions.

Continuous Learning: Actively learning emerging technologies including Rust for systems programming, LLM integration for trading signals, real-time ML inference optimization, and advanced Kubernetes operators.